

CODEQUEST: AN INTERACTIVE GAME-BASED PLATFORM FOR PROGRAMMING LEARNING

Synopsis of MSc. CS Project

Submitted in partial fulfillment of the requirements

for the award of the degree of

MASTER OF SCIENCE IN COMPUTER SCIENCE

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By

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1. INTRODUCTION

Programming is an essential skill in computer science, but beginners often find programming difficult because learning requires continuous practice, logical thinking, problem solving and the ability to understand errors. Conventional learning systems commonly present programming concepts through lessons, examples and exercises. Such approaches may not always provide an engaging environment for repeated practice.

CodeQuest: An interactive game-based platform for programming learning is proposed as an MSc Computer Science project that combines programming education with game-based learning. The system presents programming concepts through interactive challenges, progressive levels, immediate feedback, scores, badges and a leaderboard. The aim is not simply to create a game, but to design a structured learning platform in which gameplay activities are directly connected to measurable programming skill development.

The proposed platform can organize learning into beginner, intermediate and advanced challenges. A learner can select a programming track such as Python, Java or SQL, attempt challenges, receive feedback and progress through levels. The system records performance and can use the learner's previous results to recommend suitable challenges.

2. PROBLEM STATEMENT

Many learners experience difficulty in maintaining interest and confidence while learning programming. Repetitive exercises may reduce engagement, while learners may not receive immediate and meaningful feedback about their progress. A platform is therefore required that makes programming practice interactive, provides progressive challenges, records learning performance and encourages continued practice through game-based elements.

3. OBJECTIVES

- To design and develop an interactive platform for learning programming through gameplay.
- To provide programming challenges arranged according to difficulty and learning level.
- To support programming tracks such as Python, Java and SQL.
- To use points, badges, levels and leaderboards to encourage practice.
- To provide immediate feedback for submitted answers or solutions.
- To maintain learner profiles and performance history.
- To analyze learner performance and recommend appropriate challenges.
- To provide an administrator interface for managing questions, levels and content.

4. SCOPE OF THE PROJECT

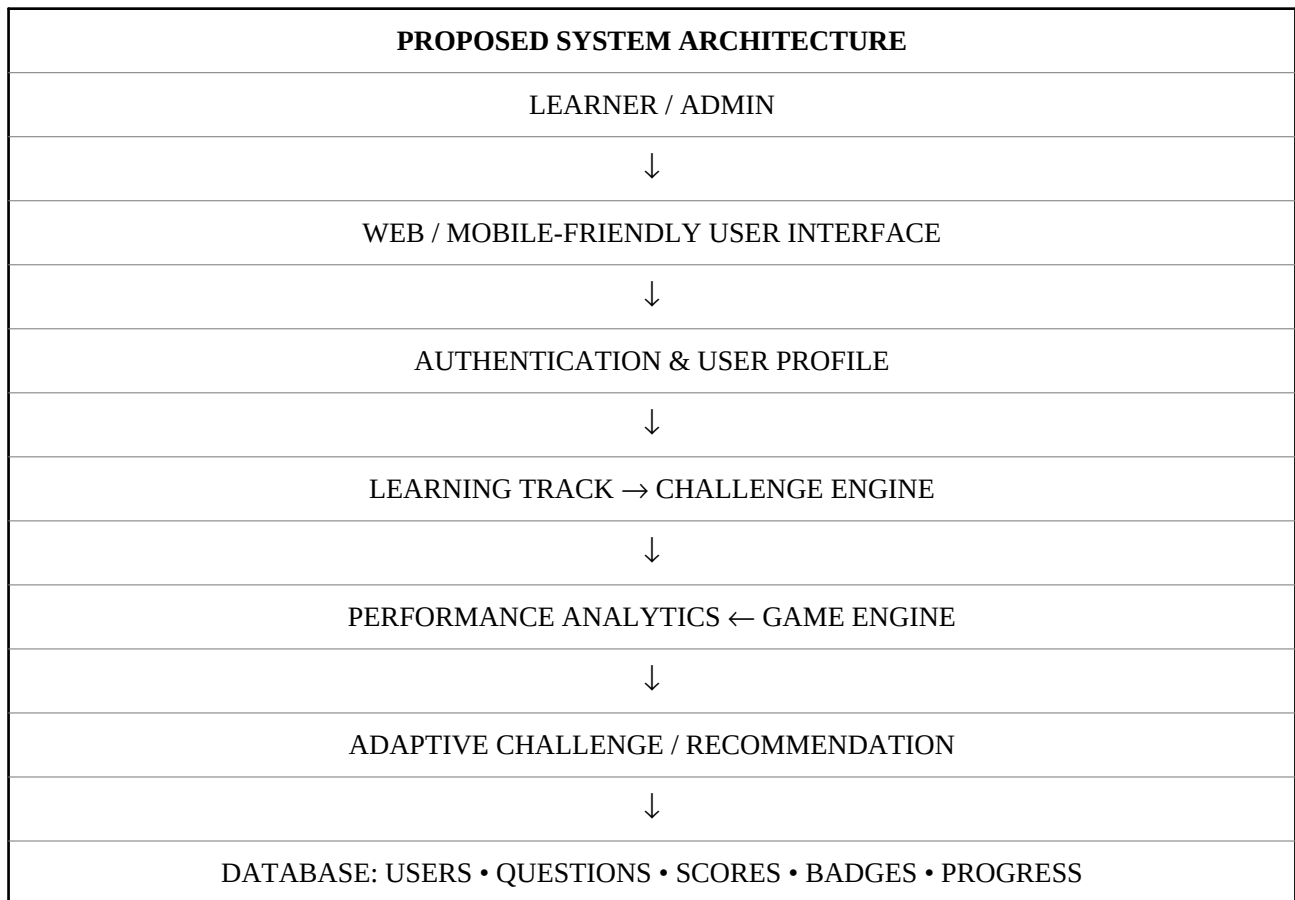
- Interactive programming challenges with beginner, intermediate and advanced levels.
- Learner registration/login, track selection and challenge attempts.
- MCQs, code-output, debugging, code-completion and logic-based challenges.
- Scores, completed levels, badges and challenge history.
- Adaptive challenge recommendation using previous performance.
- Administrator management of questions and difficulty levels.

5. EXISTING SYSTEM

Traditional programming learning generally uses textbooks, video lessons, classroom teaching, online tutorials and exercise-based practice. Several online coding-learning platforms also use interactive exercises and gamification. The proposed project is designed as an academic prototype integrating a learning path, game mechanics, performance tracking and adaptive challenge recommendation.

6. PROPOSED SYSTEM

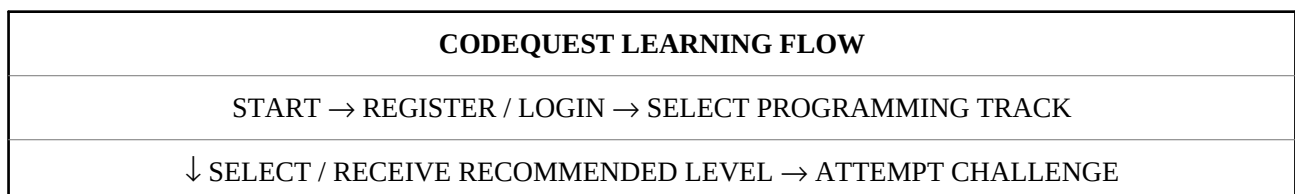
The proposed CodeQuest platform provides a structured game-based learning environment. After authentication, the learner selects a programming track and enters a sequence of challenges. Each challenge has a learning objective and difficulty level. The learner receives immediate feedback after submission. Successful performance contributes points and progress, while repeated difficulty can be used to suggest suitable practice.



7. MAJOR MODULES

Module	Main functions	Output
User Management	Registration, login, profile	Authenticated user
Learning Track	Python/Java/SQL and levels	Learning path
Challenge Engine	Coding, debugging and logic tasks	Challenge
Game Engine	Points, levels and badges	Game progress
Evaluation	Validation and feedback	Result
Performance Analytics	Scores, accuracy and weak areas	Progress report
Recommendation	Difficulty adjustment	Next challenge
Administration	Questions, levels and users	Managed platform

8. SYSTEM WORKFLOW



↓ SUBMIT → VALIDATE & GIVE IMMEDIATE FEEDBACK
↓ UPDATE SCORE + PROGRESS + BADGES → ANALYZE PERFORMANCE
↓ RECOMMEND NEXT CHALLENGE → LEVEL COMPLETED?
YES → UNLOCK NEXT LEVEL NO → PRACTICE / RETRY
↓ CONTINUE LEARNING / END

9. DATA FLOW DIAGRAM – LEVEL 0

LEVEL 0 DFD
LEARNER → [CODEQUEST SYSTEM] → SCORE / FEEDBACK / PROGRESS
↑ ↓
QUESTION BANK / DATABASE
↑
ADMIN → CONTENT & LEVEL MANAGEMENT

10. USE CASE OVERVIEW

USE CASE DIAGRAM – TEXTUAL REPRESENTATION
LEARNER: Register/Login • Select Track • Start Challenge • Submit Answer
View Feedback • Earn Points/Badges • View Leaderboard • View Progress
ADMIN: Login • Manage Users • Manage Questions • Manage Levels/Difficulty
View Performance Data

11. ADAPTIVE LEARNING COMPONENT

To give the project an MSc-level analytical component, CodeQuest can use learner performance to select the next suitable challenge. Accuracy, recent scores, completion time, attempts and difficulty can be considered. A simple recommendation model can classify current performance as improving, stable or requiring additional practice.

ADAPTIVE CHALLENGE LOGIC
PERFORMANCE DATA → ACCURACY + SCORE + ATTEMPTS + RECENT PERFORMANCE
↓ PERFORMANCE ANALYSIS → DIFFICULTY DECISION
Needs practice → Easier/reinforcement challenge
Stable → Similar-level challenge
Strong → Higher-level challenge
↓ NEXT CHALLENGE RECOMMENDATION

12. DATABASE DESIGN – MAIN ENTITIES

- User: user_id, name, login details, selected track.
- Question: question_id, track, topic, difficulty, question type and answer/solution.
- Attempt: user_id, question_id, submitted answer, result, score and timestamp.
- Progress: user_id, track, completed levels and total score.
- Badge: badge_id, badge name, condition and description.
- Recommendation: user_id, performance category, recommended difficulty and challenge.

13. TECHNOLOGY REQUIREMENTS

- Programming language: Python.
- Frontend: HTML, CSS and JavaScript.
- Backend: Python-based framework such as Flask.
- Database: SQLite or MySQL.
- Analytics/recommendation: Python data-processing or machine-learning libraries where required.
- Development environment: Windows/Linux or suitable cloud/mobile environment.

14. SYSTEM REQUIREMENTS

- Processor: Dual-core or better.
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: Minimum 10 GB free space.
- Display: Standard monitor or mobile display.
- Python 3.x and a modern web browser.
- SQLite/MySQL and a development IDE such as VS Code.

15. EXPECTED OUTCOMES

- Working prototype of an interactive programming learning platform.
- Structured programming challenges with multiple difficulty levels.
- Game-based progress using points, levels and badges.
- Immediate evaluation and feedback.
- Performance history and progress visualization.
- Adaptive/recommended challenge selection.
- Administrator facilities for maintaining programming content.

16. ADVANTAGES

- Interactive programming practice.
- Immediate feedback.
- Measurable learning progress.
- Game-based rewards and motivation.
- Multiple programming tracks.
- Scope for additional analytics and code execution.

17. FUTURE ENHANCEMENTS

- Secure online code execution for multiple languages.
- Advanced adaptive learning using machine-learning models.
- Personalized dashboards and topic-wise weakness analysis.
- Multiplayer coding competitions and collaborative challenges.
- Instructor analytics and detailed reports.
- Cloud deployment and mobile application.

18. CONCLUSION

CodeQuest proposes a focused learning environment that combines programming practice with game-based interaction. By combining challenges, immediate feedback, rewards, progress tracking and adaptive challenge recommendation, the proposed project can provide a practical and measurable approach to programming skill development.

The project involves application design, database management, gamification, performance analytics and an adaptive recommendation component, providing scope for an MSc Computer Science implementation and further enhancement.

19. REFERENCES

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